Exp: No: 11

Date: 10/9/26 Examine Network Address Translation (NAT) using Cisco Packet

Aim:

To configure and examine network Address Translation (NAT) using Cisco Packet Tracer and verify communication between a private network and a public network.

Procedure:-

Step 1:- open cisco Packet Tracer

1. Click start
2. open cisco Packet Tracer
3. Wait until the workspace loads.

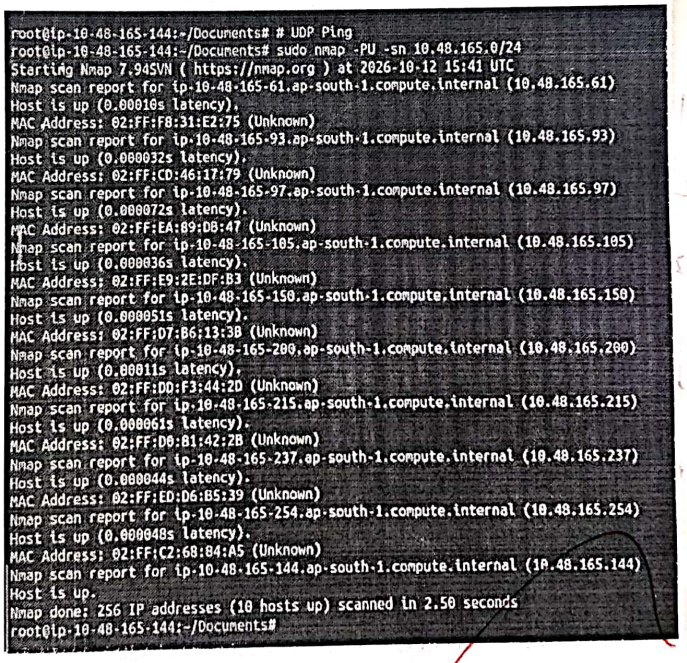
Step 2:- Add network Devices

* 1Router
* 1 switch
* 1PC
* 1server

Step 3:- connect the Devices:-

Use copper - straight Through Devices.

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```
root@ip-10-48-165-144:~/Documents# # UDP Ping
root@ip-10-48-165-144:~/Documents# sudo nmap -PU -sn 10.48.165.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-12 15:41 UTC
Nmap scan report for ip-10-48-165-61.ap-south-1.compute.internal (10.48.165.61)
Host is up (0.00010s latency).
MAC Address: 02:FF:F8:31:E2:75 (Unknown)
Nmap scan report for ip-10-48-165-93.ap-south-1.compute.internal (10.48.165.93)
Host is up (0.000032s latency).
MAC Address: 02:FF:CD:46:17:79 (Unknown)
Nmap scan report for ip-10-48-165-97.ap-south-1.compute.internal (10.48.165.97)
Host is up (0.000072s latency).
MAC Address: 02:FF:EA:89:D8:47 (Unknown)
Nmap scan report for ip-10-48-165-105.ap-south-1.compute.internal (10.48.165.105)
Host is up (0.000036s latency).
MAC Address: 02:FF:E9:2E:DF:B3 (Unknown)
Nmap scan report for ip-10-48-165-150.ap-south-1.compute.internal (10.48.165.150)
Host is up (0.000051s latency).
MAC Address: 02:FF:D7:B6:13:3B (Unknown)
Nmap scan report for ip-10-48-165-200.ap-south-1.compute.internal (10.48.165.200)
Host is up (0.00011s latency).
MAC Address: 02:FF:DD:F3:44:2D (Unknown)
Nmap scan report for ip-10-48-165-215.ap-south-1.compute.internal (10.48.165.215)
Host is up (0.000061s latency).
MAC Address: 02:FF:D0:81:42:2B (Unknown)
Nmap scan report for ip-10-48-165-237.ap-south-1.compute.internal (10.48.165.237)
Host is up (0.000044s latency).
MAC Address: 02:FF:ED:D6:B5:39 (Unknown)
Nmap scan report for ip-10-48-165-254.ap-south-1.compute.internal (10.48.165.254)
Host is up (0.000048s latency).
MAC Address: 02:FF:C2:68:84:A5 (Unknown)
Nmap scan report for ip-10-48-165-144.ap-south-1.compute.internal (10.48.165.144)
Host is up.
Nmap done: 256 IP addresses (10 hosts up) scanned in 2.50 seconds
root@ip-10-48-165-144:~/Documents#
```

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Configure the Router

Step 4:- Open Router CLI

Step 5:- Configure the Inside Interface

enable

Configure terminal

interface gigabit Ethernet 0/0

ip address

ip nat inside

no shutdown

exit

Step 6:- Configure the outside interface

interface gigabit Ethernet 0/1

ip address

ip nat outside

no shut down

exit.

Step 7:- Configure static NAT

Router (config) # ip nat inside source

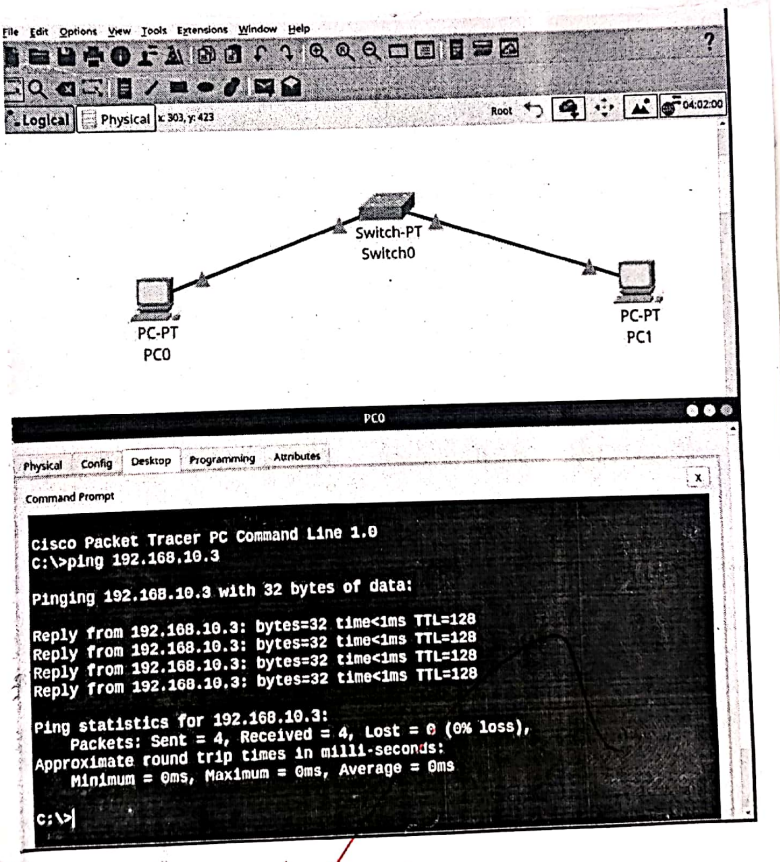
static 192.168.12.200 1.1.1.100

Step 8:- Save the configuration

end

Press Enter.

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Configure the PC

Step 9:- Assign IP Address

1. click PC0
2. select Desktop
3. Enter.

Configure the server

Step 10:- Assign IP Address

1. Click Server.
2. select Desktop
3. Enter

Verify NAT

Step 11:- Test Connectivity

Step 12:- verify NAT Translation

Expected output:-

Pro	Inside global	Inside local
	200.1.1.100	192.168.1.2

Step 13:- View NAT Statistics.

Result:- Network Address Translation (NAT) was successfully examined using cisco packet tracer. NAT Translated the private IP adarers, enabling communication with external networks. The connectivity was verified using the Ping command